K. J. Somaiya School of Engineering, Mumbai-77

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Department of Computer Engineering

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Course Name: Web Development Laboratory

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Experiment No: 4

Title

Form Validation using jQuery for a Climate Intelligence and Heatwave Early Warning Web Portal

Aim of the Experiment

To design and implement client-side form validation using jQuery for a climate monitoring and heatwave alert registration form, ensuring that user-entered data such as name, mobile number, email, location, temperature, humidity, alert level, monitoring date, and notification preference is complete, valid, and meaningful before form submission.

Objectives of the Experiment

- To create an interactive HTML form for collecting user and climate-monitoring information.
- To use jQuery to validate text, numeric, email, date, and selection fields.
- To implement regular expressions for validating mobile numbers, email addresses, and location/station IDs.
- To use jQuery selectors, events, DOM manipulation, and class-based styling to display validation messages.
- To prevent form submission when invalid or incomplete data is entered.
- To apply form validation to a real-world climate intelligence and heatwave early-warning use case.

COs to be achieved

CO1: Developing webpages using HTML, CSS, JavaScript and jQuery.

Books / Journals / Websites references

1. jQuery API Documentation – Selectors, Events and DOM Manipulation.

2. MDN Web Docs – HTML Forms and JavaScript Form Validation.

3. W3Schools – jQuery Form Validation and Event Handling.

4. JavaScript and jQuery documentation/reference material prescribed by the course instructor.

Theory

Form validation is the process of checking whether the information entered by a user satisfies predefined rules before it is submitted or processed. Validation improves data quality, usability, and reliability of web applications.

jQuery is a JavaScript library that simplifies DOM selection, event handling, and manipulation. In form validation, jQuery selectors can identify form controls, while methods such as val(), trim(), addClass(), removeClass(), text(), and event handlers such as submit() and input() can be used to validate and provide immediate feedback.

Common validation rules include checking for empty fields, minimum and maximum length, valid email format, valid mobile number, acceptable numeric ranges, selected options, and valid dates. Regular expressions are useful for validating structured values such as mobile numbers, email addresses, and station identifiers.

In this experiment, jQuery validation is applied to a Climate Intelligence and Heatwave Early Warning Web Portal. The portal collects location and climate-observation information that can be used to display heatwave status and early-warning notifications.

Problem Statement

"jQuery Form Validation for Climate Monitoring and Heatwave Early Warning"

Develop a web-based climate monitoring registration form for the Climate Intelligence and Heatwave Early Warning Web Portal. The form should collect essential user and climate-observation information and use jQuery to validate the entered values before submission. Invalid or incomplete data must be identified and appropriate error messages must be displayed to the user.

Task 1: Create the Climate Monitoring Registration Form

Scenario

Design a registration and climate-observation form for users who will monitor local temperature conditions and receive heatwave early-warning notifications.

Requirements

- Create fields for Observer Name, Mobile Number, Email, Location/Area, Weather Station ID, Temperature (°C), Relative Humidity (%), Heatwave Alert Level, Monitoring Date, and Notification Preference.
- Use <form>, <label>, <input>, <select>, <option>, <fieldset> and <legend>.
- Use suitable input types such as text, email, number, date, and radio buttons.
- Provide Submit and Reset buttons.
- Assign unique id and name attributes to all fields.

Concepts Tested

- HTML Forms
- Form controls
- jQuery selectors
- DOM element identification

Deliverable: climate-alert.html

Task 2: Validate Required Text Fields

Scenario

Ensure that essential observer and location information is not left blank.

Requirements

- Check that Observer Name, Location/Area and Weather Station ID are not empty.
- Remove leading and trailing spaces using jQuery trim-related validation logic.
- Validate that Observer Name contains alphabetic characters and spaces only.
- Validate the Weather Station ID using a suitable regular expression.
- Display an error message beside each invalid field.

Concepts Tested

- jQuery selectors
- Conditions
- String validation
- DOM manipulation
- Required-field validation

Deliverable: climate-alert.html

Task 3: Validate Mobile Number, Email and Climate Values

Scenario

Verify that contact information and observed climate values follow the expected format and range.

Requirements

- Validate a 10-digit Indian mobile number using a suitable regular expression.
- Validate the email address using a suitable regular expression or HTML input validation.
- Validate Temperature (°C) within a meaningful input range, such as -10°C to 60°C.
- Validate Relative Humidity (%) within the range 0 to 100.
- Display specific error messages for invalid mobile number, email, temperature, or humidity.
- Do not allow the form to submit until all required values are valid.

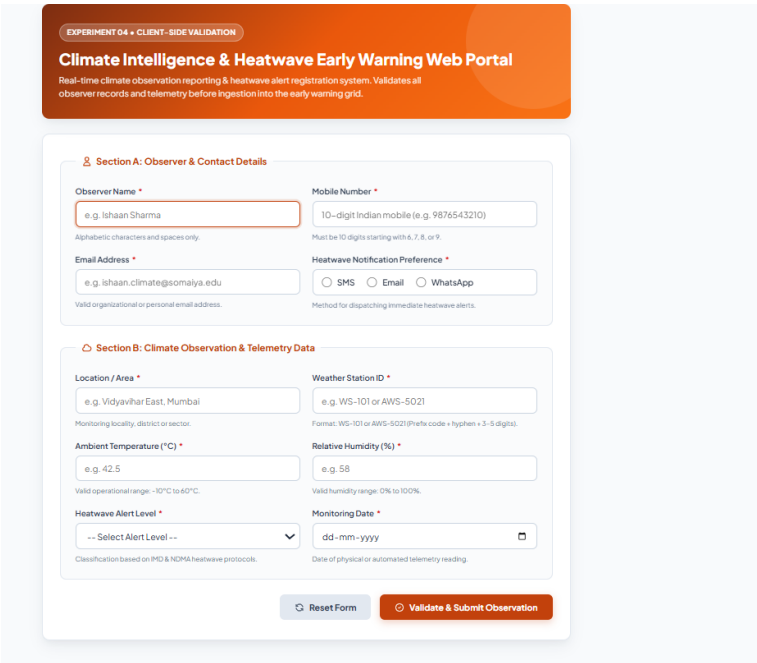
Concepts Tested

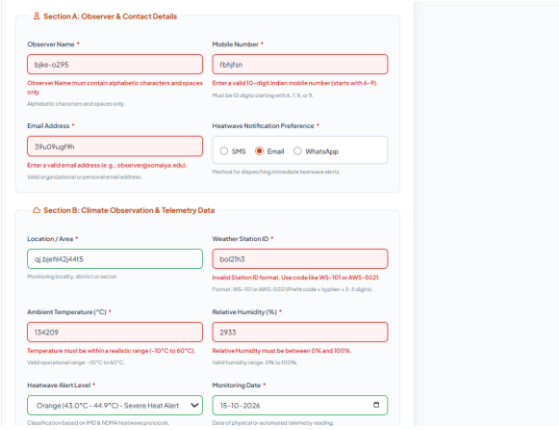
- jQuery selectors
- Regular expressions
- Numeric validation
- String validation
- Form submit event

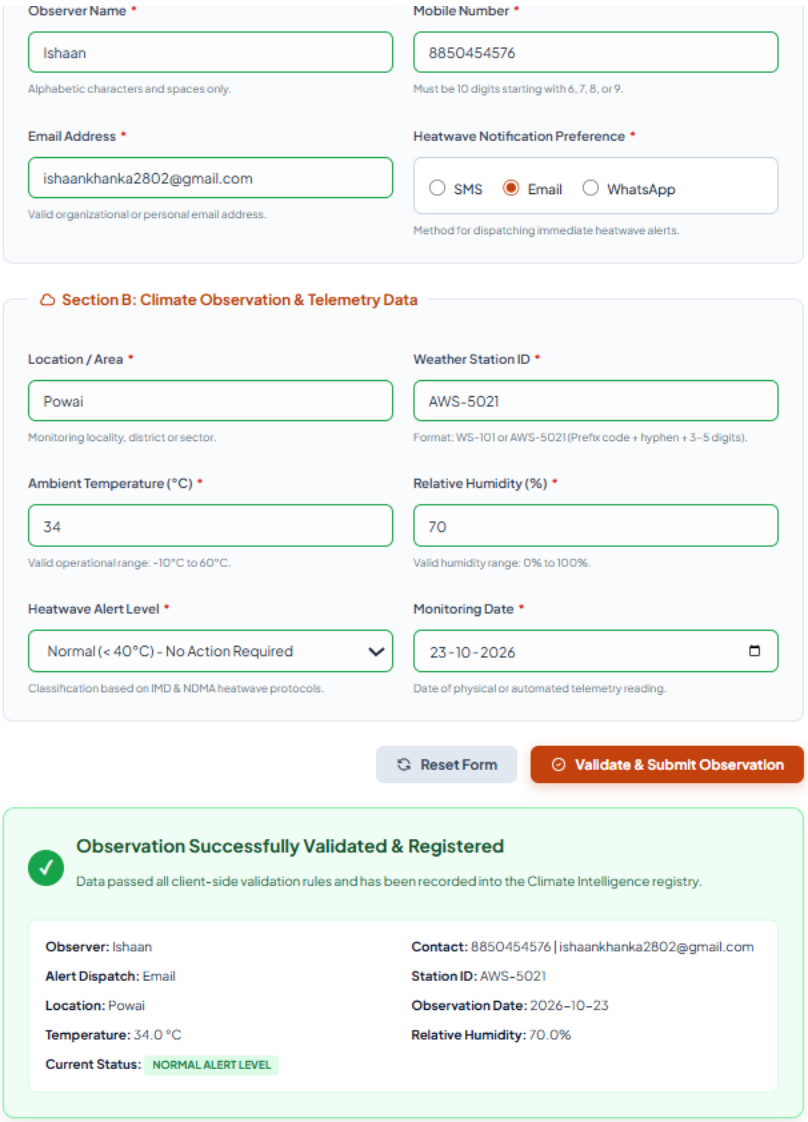
Deliverable: climate-alert.html

Output / Screenshot:

Students should execute climate-alert.html in a web browser and paste screenshots showing: (1) the climate monitoring registration form, (2) validation errors for invalid inputs, and (3) the success message after entering valid data.







Conclusion and Discussion:

The experiment demonstrates how jQuery can be used to validate user input on a web form before submission. Validation rules were applied to observer details, contact information, location and weather-station details, temperature, humidity, alert level, monitoring date, and notification preference.

Discussion: Students should test the form with both valid and invalid inputs, record the observed behavior, and take feedback from fellow classmates and faculty. Students should also observe how jQuery selectors, event handling, regular expressions, numeric comparisons, and DOM

manipulation contribute to an interactive climate intelligence and heatwave early-warning web portal.